The Correlation Coefficient

2026-04-17

Introduction

The correlation coefficient rescales covariance into a unitless number between -1 and $+1$, making comparisons across different measurement scales meaningful. A correlation of $+1$ means perfect positive linear dependence, -1 perfect negative, and 0 no linear association. It is the single most reported bivariate summary in applied research.

Prerequisites

Variance and covariance.

Theory

For random variables X, Y with finite non-zero variances,

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Cauchy-Schwarz guarantees $\rho \in [-1, 1]$. Equality at ± 1 occurs if and only if $Y = aX + b$ almost surely with $\mathrm{sign}(a) = \mathrm{sign}(\rho)$.

The sample correlation coefficient (Pearson's r) is the plug-in estimator using sample covariance and SDs:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}.$$

Key caveats:

- $\rho = 0$ does **not** imply independence unless (X, Y) is jointly normal.
- ρ measures only the *linear* component of association; non-linear dependence is invisible to it.
- Outliers affect r disproportionately; a robust alternative is the Spearman rank correlation.

Assumptions

- Both variances finite and non-zero.
- Linearity is the interpretive assumption: ρ captures only linear dependence.

R Implementation

```
library(mvtnorm)
set.seed(2026)

# Simulate bivariate normals at several correlation levels
rho_grid <- c(-0.8, -0.3, 0, 0.3, 0.8)
n <- 500
results <- sapply(rho_grid, function(rho) {
```

```r
  Sigma <- matrix(c(1, rho, rho, 1), 2, 2)
  XY <- rmvnorm(n, sigma = Sigma)
  cor(XY[, 1], XY[, 2])
})
data.frame(true = rho_grid, empirical = round(results, 3))

# Zero correlation does not imply independence
X <- rnorm(1e4)
Y <- X^2 - 1      # quadratic, mean zero
cor(X, Y)

# Outlier sensitivity
X <- rnorm(100); Y <- 0.3 * X + rnorm(100)
c(no_outlier = cor(X, Y),
  with_outlier = cor(c(X, 20), c(Y, -20)))
```

Output & Results

```
  true  empirical
1 -0.8     -0.790
2 -0.3     -0.307
3  0.0     -0.023
4  0.3      0.309
5  0.8      0.815

[1] 0.005          # zero Pearson despite clear dependence

no_outlier with_outlier
     0.309       -0.059
```

Sample correlations recover the population values within Monte Carlo error. The $X, X^2 - 1$ example shows zero correlation with obvious dependence. A single outlying point flips the sign of a small-sample correlation.

Interpretation

Reported in a paper as "r = .68 (95% CI .56-.77), $p < .001$", the correlation conveys both direction and magnitude of linear association. Always accompany r with a scatter plot to verify linearity; never report r alone.

Practical Tips

- Use Pearson only for roughly linear relationships between continuous variables.
- For monotone non-linear association, use Spearman's rho or Kendall's tau.
- A 95% CI for ρ uses the Fisher z-transformation: $\text{atanh}(r)$ has an approximately normal sampling distribution.
- Correlation does not imply causation; confounders can create strong associations between unrelated variables.
- In multivariate data, a single correlation matrix can be visualised with `corrplot::corrplot()` or `ggcorrplot::ggcorrplot()`; always plot the matrix and the pairwise scatters.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**

- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots